Autistic adults more often report absent sensory imagery: a preregistered replication with an ADHD comparison

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Abstract

Arnold et al. (2026) found that autistic adults report weaker mental imagery than non-autistic adults and are more likely to report no smell or taste imagery. Following a registered plan, we tested these findings in a separate online survey of young UK adults (New et al., 2026). We compared 150 respondents who reported diagnosed or strongly suspected autism with 749 reporting no neurodevelopmental condition and 84 reporting attention-deficit/hyperactivity disorder (ADHD) without autism. All three predictions were supported. Imagery in the autism group was 0.42 standard deviations lower on average, and the odds of reporting no smell or taste imagery were 2.7 and 3.8 times higher. The difference was no larger for smell and taste than for other senses, and not significantly larger in autism than in ADHD. Further, unregistered analyses asked whether this lower average reflected the whole autism group or a subgroup. In New et al.'s survey, 20% of the autism group reported no imagery in at least one sense, versus 6.5% of the comparison group, and a simple shift or wider spread of scores could not reproduce this. In Arnold et al.'s data, absent imagery was also more common in autism (9.8% vs 4.6%), but mostly because autistic adults' scores were lower and more spread out. Because much of the difference comes from a minority who report no imagery, studies should report the percentage of people with absent imagery alongside average scores, in order to tell whether autism involves weaker imagery overall or absent imagery in some people.

Keywords: autism; mental imagery; aphantasia; ADHD; replication; preregistration; quantile regression

Highlights

• Preregistered test replicates weaker imagery in autism in a new sample

• Odds of no smell or taste imagery were 2.7 and 3.8 times higher in autism

• The difference was no larger for smell and taste than for other senses

• In the new sample, a minority with no imagery carried most of the difference

• A simple shift or wider spread of scores could not explain this pattern

Introduction

Try to imagine the smell of fresh coffee, or the face of a close friend. Most people report some sensory experience when they do this, although how vivid it is varies a great deal from person to person. A minority report no experience at all. When this applies to visual imagery it is called aphantasia (Zeman et al., 2015), and many people with aphantasia also report weak or no imagery in other senses, such as sound or smell (Dance, Ward & Simner, 2021). In this paper, "imagery" means what people say about this kind of experience on questionnaires. We do not measure how well people can use imagery in a task.

Several recent studies link autism with weaker reported imagery. Arnold et al. (2026) surveyed 327 adults who reported a formal autism diagnosis, using the Plymouth Sensory Imagery Questionnaire. Compared with a non-autistic group, they reported weaker imagery across the senses, and more of them reported no smell or taste imagery at all. King et al. (2024) found lower visual imagery in a group that included both diagnosed and self-identifying autistic adults, and more of that group scored low enough to count as aphantasic. Looking at the link from the other direction, Dance, Jaquiery et al. (2021) found that people with aphantasia reported more autistic traits than people without it. Two studies complicate this picture. Bouyer et al. (2025) pointed out that the Autism Spectrum Quotient, the most widely used questionnaire for autistic traits, includes questions about imagination. Correlating it with imagery therefore partly measures imagery twice, and once those questions were set aside, the link they found was weak. Taylor et al. (2026) found that lower imagery went with more autistic traits, but that imagery was barely related to sensory sensitivity.

Most of this evidence compares the average scores of two groups. A lower average, however, can come about in quite different ways. The whole autistic group might score slightly lower than the comparison group. The autistic group might score lower on average and also vary more from person to person. Or most autistic people might report imagery just like

anyone else, while a minority report none at all in one or more senses. Only the third pattern would link autism to the complete absence of imagery in some people, and so to aphantasia. It would also mean that an average difference is misleading in both directions. It understates how different the minority with absent imagery is, and it overstates how different everyone else is. Arnold et al.'s finding that more autistic adults report no smell or taste imagery hints at the third pattern, but comparing averages cannot tell the three apart.

New et al. (2026) surveyed mental imagery in 1,003 young UK adults. Their survey also asked whether respondents had been diagnosed with, or strongly suspected, a neurodevelopmental condition, and which one. New et al. analysed all of these respondents as one group. Separating them allows three things. First, we can test Arnold et al.'s three main findings in a different sample, with a different questionnaire. We registered this test before running it. Second, attention-deficit/hyperactivity disorder (ADHD) was reported almost as often as autism. This lets us check whether the imagery difference is specific to autism or appears with any neurodevelopmental condition. If it appears with any such condition, the ADHD group should show it too. Third, we can see where in the range of scores the autism difference lies. We decided to do this third analysis only after seeing the registered results, so we report it separately as exploratory, meaning that it can suggest a pattern but cannot confirm one. We repeated it in Arnold et al.'s public data. Because those data had inspired our predictions and we had analysed them before registering, this repeat is a consistency check, not an independent test.

Methods

Data and registration

We reanalysed Study 2 of New et al. (2026), an online survey of mental imagery in young UK adults. The data are freely available on the Open Science Framework (OSF; project 2bdn6). New et al. recruited participants through Prolific, a website for recruiting research volunteers, and advertised the study as a reading study without mentioning imagery. To take part, people had to live in or be nationals of the UK, speak English as a first language, and be aged 18 to 30.

We registered our analysis plan on OSF on 10 October 2026, before running any comparison involving autism or ADHD (https://osf.io/87udm). Before registering, we had read both source papers, reanalysed Arnold et al.'s public data, and seen New et al.'s result for the combined neurodevelopmental group. The registration lists this prior knowledge. The registered analysis was run once, on the day of registration, and we call its results "registered". Analyses we added after seeing those results are labelled "exploratory" throughout. Most of them were run on 10 and 11 October. Two were added later, after an internal review: a check that drops one sense from both questionnaires (17 October) and the shift benchmark described below (18 October). Every departure from the plan is listed in Table S8. The most important is an error in how the plan matched the two questionnaires, described under "Comparison with Arnold et al.".

Sample size and transparency

The number of participants was fixed by the existing data, so we could not choose it. Before registering, we calculated the chance that our tests would detect a real difference (statistical power), based on the size of the differences in Arnold et al.'s data and the group sizes in New et al.'s survey. For the overall imagery score, power was 79% if the true difference was as large as Arnold et al.'s and 34% if it was half that size. For absent smell and taste imagery, assuming 2.4% of the comparison group reported no imagery, power was 87% if the true odds ratio was 3 and 46% if it was 2. We report how we determined our sample size, all data exclusions (if any), all inclusion/exclusion criteria, whether inclusion/exclusion criteria were established prior to data analysis, all manipulations, and all measures in the study. The only exclusions were New et al.'s attention-check exclusions and the 20 respondents who reported only other conditions, and both were set out in the registered plan. The study had no experimental manipulations. Every measure we analysed is reported in the main text or in Table S7.

Ethics

[CONFIRM OR REPLACE] This study is a secondary analysis of existing survey data that are publicly archived and contain no names or contact details. No participants were contacted, no identifying information was used, and only group-level results are reported. Participants in the original studies gave informed consent. New et al.'s consent form stated that the anonymous data would be used in future research and could be shared with other researchers. The original data collections were approved by [New et al.: ethics committee and approval number] and [Arnold et al.: ethics committee and approval number]. [State here whether a separate ethics review was sought for this secondary analysis.]

Participants

New et al. removed respondents who failed an attention check. Respondents picked their five favourite activities from a list twice during the survey, and anyone who did not repeat at least three of them was removed. This left 1,003 of the 1,020 respondents.

One question asked whether the respondent had been diagnosed with, or strongly suspected, a neurodevelopmental condition. Those who answered yes then ticked the conditions that applied. We formed three groups from these answers.

- **Autism group** (150 people): ticked autism. Forty-six of them also ticked ADHD.
- **ADHD-only group** (84 people): ticked ADHD but not autism.
- **Comparison group** (749 people): reported no neurodevelopmental condition.

We left out 20 respondents who ticked only other conditions. The groups are based on what people reported about themselves, not on a clinical assessment. One person in the comparison group had no recorded age, so the analyses that adjust for age use 898 people when comparing autism with the comparison group, and 982 when all three groups are included. Table 1 describes the groups.

Table 1. Participants in New et al.'s survey

	Comparison	Autism	ADHD-only
People	749	150	84
Also reported ADHD	-	46	-
Female	49%	57%	54%
Age, mean (SD)	25.1 (3.6)	25.0 (3.4)	25.2 (3.4)
Overall imagery, mean (SD)	+0.00 (1.00)	-0.43 (1.40)	-0.07 (1.16)
Absent imagery in at least one sense	6.5%	20.0%	8.3%

Overall imagery is measured in comparison-group standard deviations, so the comparison group has an average of 0 and a standard deviation (SD) of 1; negative values mean weaker imagery than the comparison group's average. A larger SD means scores are more spread out. Absent imagery means the lowest possible score in a sense. Sex was recorded as female, male or other; two respondents in these groups chose other. Ages ranged from 18 to 32: New et al.'s eligibility was 18 to 30, but three respondents gave older ages. One comparison respondent had no recorded age. Table S1 gives the raw score for each sense.

How imagery was measured

New et al. used an extended version of the Questionnaire upon Mental Imagery (QMI; Sheehan, 1967). Respondents imagined familiar things, such as a friend's face, a particular sound or a particular taste, and rated how vivid the experience was on a scale from 1 to 7. A rating of 1 means that they could think about the item but had no sensory experience of it. A rating of 7 means the experience was as vivid as the real thing. Each sense has five items, so each sense score runs from 5 to 35.

New et al. measured nine kinds of imagery. We used the six that our registered plan matched to the six senses in Arnold et al.'s questionnaire: sight, sound, smell, taste, touch and movement. Results for the other three (emotion, voice and sensations inside the body) are in Table S7.

A score of 5 in a sense means the person chose "no experience" for all five items in that sense. We call this **absent imagery** in that sense. It is the strictest possible definition, it matches the definition in New et al.'s own analysis code, and it is the questionnaire equivalent of reporting no image at all, as people with aphantasia do. It describes what people reported, not a test of what they can do.

The **overall imagery score** combines the six senses. Because the senses differ in how spread out their scores are, we first put each sense on a common scale. A standard deviation measures how spread out scores are around the average; in a typical group, about two-thirds of people score within one standard deviation of the average. We expressed each person's score in each sense as the number of comparison-group standard deviations above or below the comparison group's average. We then averaged the six senses and rescaled the result in the same way. As a result, the comparison group has an average of 0 and a standard deviation of 1, each sense counts equally, and a group difference of -0.4 means the autism group scored on average 0.4 of a comparison-group standard deviation lower.

Registered analyses

Main tests. We tested three predictions from Arnold et al., in a fixed order.

1. The autism group has a lower overall imagery score.
2. Absent smell imagery is more common in the autism group.
3. Absent taste imagery is more common in the autism group.

Arnold et al.'s results told us which direction to expect, so each test looked only for a difference in that direction (a one-sided test), using the usual 5% significance level. A later prediction could count as supported only if all earlier ones were. Testing in a fixed order like this keeps the overall chance of a false positive across the three tests at 5%, without any further correction. For each result we also report a 95% confidence interval, the range of values that are compatible with the data.

For the overall score we used linear regression, which estimates the difference in average scores between groups. We used a type of standard error (HC3) that stays accurate even when one group's scores are more spread out than the other's. For absent smell and taste imagery we used logistic regression, which estimates how the odds of an outcome differ between groups. An odds ratio of 2, for example, means that the odds of absent imagery were twice as high in the autism group. All models adjusted for age and sex, so that differences in these between the groups do not distort the comparison.

Two people in the analysed groups described their sex as "other", and neither reported absent smell or taste imagery. With so few people in a category, ordinary logistic regression cannot produce an estimate. As our plan allowed, we therefore used Firth's version of logistic regression (Firth, 1993), which still gives sensible estimates in this situation. As registered, its confidence intervals are of the simplest kind (Wald intervals), which can be less accurate when a category is very small.

Secondary tests. Five further tests asked whether the autism difference was specific to particular senses or to autism.

• S1(a): Is the autism difference larger for smell and taste than for sight?

• S1(b): Does the size of the autism difference vary at all across the six senses?

• S2: Is the overall difference larger for autism than for ADHD-only?

• S3: Is absent visual imagery more common in the autism group?

• S4: Does the ADHD-only group show a larger difference for smell and taste than for sight?

Running several tests raises the chance that at least one is significant by luck alone. We corrected for this with Holm's (1979) method, which makes the significance threshold stricter for each additional test. For S1 and S4, each person contributed six scores, one per sense, and the analysis took into account that scores from the same person are related.

Sensitivity analyses. To check that the main result did not depend on particular choices, we repeated the overall-score test four ways: leaving out autistic respondents who also reported ADHD; using the original 5-35 scores instead of the rescaled ones; including only people aged 18 to 30 (three respondents were older); and without adjusting for age and sex.

Comparison with Arnold et al.

Arnold et al. (2026) published imagery scores for 327 adults who reported a formal autism diagnosis and 327 comparison adults. They used the Plymouth Sensory Imagery Questionnaire (PSI-Q; Andrade et al., 2014), on which each sense is scored from 0 (no image at all) to 50. We ran the same analyses on all 327 autistic adults. Arnold et al.'s own analysis kept only the 242 who scored 26 or more on the Autism Spectrum Quotient (AQ), a questionnaire measuring autistic traits. New et al.'s survey did not include the AQ, so the full group of 327 is the closer match to our autism group.

Our registered plan assumed that Arnold et al.'s sixth sense was movement, like New et al.'s. It is in fact bodily sensation, such as feeling warmth or a sore throat. We found this mistake while preparing the manuscript, after the registered results were known. We kept the registered results as they were run. To check whether the mismatch affected them, we added one analysis that leaves out the sixth sense in both datasets and uses only sight, sound, smell,

taste and touch. We recorded this check before running it, but after seeing the registered results, so it counts as exploratory.

Exploratory analyses: where is the difference?

These analyses ask why the autism group's average was lower. One possibility is that the whole group scored a little lower. Another is that autistic respondents used the ends of the rating scale more often. A third is that a smaller subgroup reported no imagery at all. We ran each analysis in both datasets. In Arnold et al.'s data, a score of 0 means absent imagery and 50 is the top score.

- **Ends of the scale.** For each sense, we counted how many people gave the lowest possible score (absent imagery) and how many gave the highest. If autistic respondents simply used the ends of the scale more, both extremes should be more common. We compared these percentages between groups with Fisher's exact test, a standard test for comparing percentages that works well with small numbers. We also counted, for each person, in how many senses they gave the lowest score and in how many the highest.
- **Percentiles.** The 10th percentile is the score below which the lowest-scoring 10% of a group fall; the 50th percentile, or median, is the middle score. We compared the groups at the 10th, 25th, 50th, 75th and 90th percentiles of the overall score, using quantile regression (Koenker & Bassett, 1978), which compares groups at a chosen percentile instead of at the average, adjusted for age and sex. If the whole autism group scored a little lower, the difference should be similar at every percentile. If a subgroup reported no imagery, the difference should be large at the low percentiles and small elsewhere.
- **People with imagery in every sense.** We repeated the overall comparison after leaving out everyone with absent imagery in any sense. This removes more autistic than comparison respondents, because absent imagery is more common in autism, so this comparison on its own could be misleading. We therefore read it together with the shift benchmark.

- **Shift benchmark.** A large difference at low percentiles could also appear without any separate subgroup, if autistic respondents' scores were simply lower and more spread out than the comparison group's. To test this, we built two imaginary autism groups from the comparison group's own answers. In the "shift" version, we lowered every comparison respondent's score in each sense by the average autism-comparison difference in that sense. In the "shift + stretch" version, we also spread out each sense's scores until they were as variable as the autism group's. In both versions, scores were rounded to whole points and kept within the questionnaire's range, and each person kept their own pattern across the senses. We then compared these imaginary groups with the real autism group on the measures above. If either imaginary group looked like the real one, a simple shift or spread would be enough to explain the data, and there would be no need for a separate subgroup. To judge how precise the real group's values were, we repeatedly resampled the autism group (2,000 times) and report the middle 95% of the results (a bootstrap interval). All values in this comparison are unadjusted.

We also ran several other exploratory analyses. These covered the most attentive respondents only, the three kinds of imagery not in the overall score, and additional adjustment for current treatment, education and a self-rating of empathy. They also covered Arnold et al.'s AQ-selected subgroup and a reanalysis of public data from Dance, Ward and Simner (2021). Table S7 gives every result and the reason for each analysis.

Reproducibility

The analysis code and all group-level results are available at https://github.com/i-habib/aphantasia/tree/main/projects/P5_asd_adhd_replication. We do not redistribute the original data, which are available from their original archives. We reproduced all registered results in a separate run, and checked the Firth models against an independent implementation in the R package brglm2. Analysis code was written and checked with the help of AI coding assistants, as described in the declaration on generative AI. Every registered result was reproduced independently of the original run.

Results

Registered main tests

All three main predictions were supported (Table 2, Figure 1). The autism group's overall imagery score was 0.42 standard deviations lower than the comparison group's (95% CI -0.66 to -0.19; one-sided p = .0002). By common conventions this is a small-to-moderate difference.

Absent smell imagery was reported by 15 of 150 autistic respondents (10.0%) and 29 of 749 comparison respondents (3.9%). After adjusting for age and sex, the odds of absent smell imagery were 2.69 times higher in the autism group (95% CI 1.42 to 5.10; one-sided p = .001). Absent taste imagery was reported by 17 autistic respondents (11.3%) and 23 comparison respondents (3.1%), and the odds were 3.81 times higher in the autism group (95% CI 2.00 to 7.26; one-sided p < .001).

Table 2. Registered tests in New et al.'s survey

Test	Estimate (95% CI)	p (main tests)	p corrected for multiple tests	Supported
H1. Overall imagery, autism vs comparison	-0.42 SD (-0.66 to -0.19)	< .001	-	Yes
H2a. Absent smell imagery	OR 2.69 (1.42 to 5.10)	.001	-	Yes
H2b. Absent taste imagery	OR 3.81 (2.00 to 7.26)	< .001	-	Yes
S1(a). Autism: smell/taste difference minus sight difference	+0.03 SD (-0.15 to +0.20)	-	1.00	No
S1(b). Autism: difference varies across the six senses	chi-square 4.34, 5 df	-	1.00	No
S2. Autism minus ADHD-only, overall score	-0.36 SD (-0.69 to -0.02)	-	.149	No
S3. Absent visual imagery, autism vs comparison	OR 2.84 (1.27 to 6.33)	-	.054	No
S4. ADHD-only: smell/taste difference minus sight difference	+0.02 SD (-0.15 to +0.20)	-	1.00	No

H1 to H2b are the three main tests. They were run in a fixed order and each looked for a difference in the predicted direction only (one-sided p). S1 to S4 are the secondary tests. Their p values are corrected for running five tests (Holm's method), so a test can have a 95% confidence interval that excludes zero, as S2 does, and still not be significant after correction. All models adjust for age and sex. SD: difference in comparison-group standard deviations. OR: odds ratio, how many times higher the odds of absent imagery were in the autism group. Number of people: 898 for H1 to H2b, S1 and S3; 982 for S2; 832 for S4.

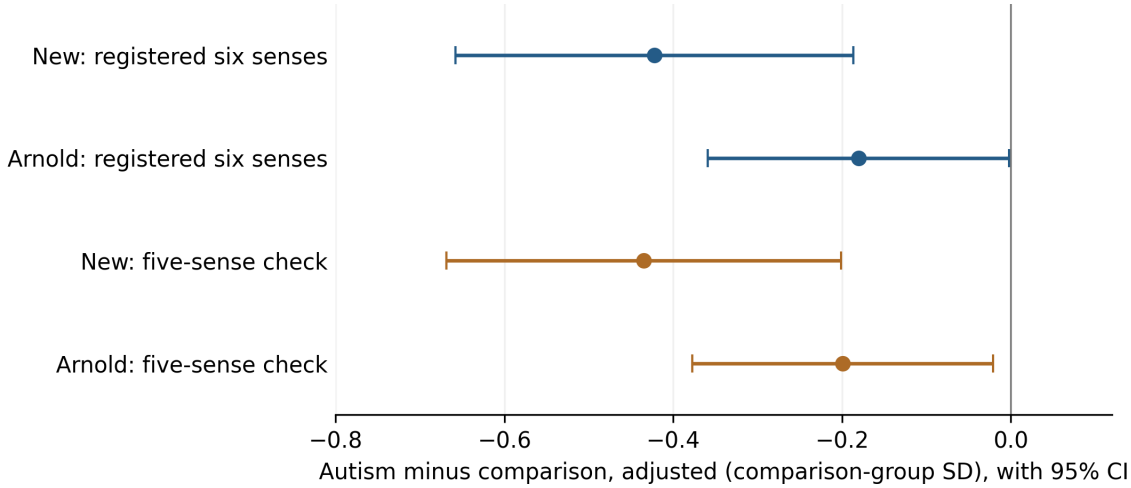


Figure 1. Difference in overall imagery between the autism and comparison groups, adjusted for age and sex. Each point is the estimated difference and each line its 95% confidence interval. Negative values mean the autism group reported weaker imagery. Blue points use the registered score based on six senses. Orange points use only the five senses that both questionnaires measure in the same way (an exploratory check). Each dataset is measured against its own comparison group. The four rows come from two datasets, not four independent samples.

Registered secondary tests

None of the five secondary tests was statistically significant once we corrected for the number of tests (Table 2).

The autism difference was not larger for smell and taste than for sight (S1a: +0.03 standard deviations, 95% CI -0.15 to +0.20), and the size of the difference did not vary detectably across the six senses (S1b: chi-square 4.34 with 5 degrees of freedom, p = .50). So although absent smell and taste imagery was more common in the autism group, the average gap in those two senses was no bigger than in the other senses.

The autism group's overall score was 0.36 standard deviations below the ADHD-only group's (95% CI -0.69 to -0.02). On its own this difference had p = .037, but after correcting for the number of tests it was not statistically significant (p = .15). The registered test of whether the difference is specific to autism was therefore not supported. The ADHD-only

group differed little from the comparison group (-0.07, 95% CI -0.32 to +0.19).

Absent visual imagery was more common in the autism group (odds ratio 2.84, 95% CI 1.27 to 6.33), but this was just short of statistical significance after correcting for the number of tests (p = .054). The ADHD-only group showed no special pattern for smell and taste (S4: +0.02, 95% CI -0.15 to +0.20).

Sensitivity analyses

The overall difference stayed much the same in all four checks. It was -0.39 without autistic respondents who also reported ADHD (95% CI -0.65 to -0.13), -2.2 points on the original 5-35 scale (95% CI -3.5 to -1.0), -0.42 when only people aged 18 to 30 were included, and -0.43 without adjusting for age and sex. Table S4 gives all sensitivity results.

Comparison with Arnold et al. and the five-sense check

In Arnold et al.'s full sample, the registered six-sense difference was smaller, -0.18 standard deviations (95% CI -0.36 to -0.002). Leaving out the mismatched sixth sense made little difference in either dataset (Figure 1). Using five senses, the difference was -0.43 (95% CI -0.67 to -0.20) in New et al.'s data and -0.20 (95% CI -0.38 to -0.02) in Arnold et al.'s.

Exploratory results: where is the difference?

In both datasets, the difference was concentrated among the lowest scorers (Table 3, Figure 2). Whether this reflects a separate subgroup, rather than a lower and more spread-out group, differed between the datasets (Table 4).

New et al.'s survey. Twenty percent of autistic respondents reported absent imagery in at least one of the six senses, compared with 6.5% of the comparison group (Fisher's exact test, p < .001). Absent imagery was more common in the autism group in every sense, but top scores were about equally common in both groups (Table S6). On average, autistic respondents had absent imagery in 0.34 more senses (95% CI 0.14 to 0.54) and the same number of top-scoring senses (difference -0.03, 95% CI -0.21 to +0.15). So the autism group was not simply more

likely to choose the extremes of the scale.

The size of the difference depended on where in the range of scores we looked. Among the lowest-scoring 10% of each group (the 10th percentile), the autism group scored 1.27 standard deviations lower (95% CI -1.66 to -0.88). At the 25th percentile the difference was 0.56 (95% CI -0.81 to -0.30). From the median upwards it was between 0.09 and 0.19, and none of these differences was statistically significant. Among the 819 respondents who reported some imagery in every sense, the autism and comparison groups differed by only 0.03 standard deviations (95% CI -0.21 to +0.14).

Neither imaginary group from the shift benchmark could reproduce this pattern (Table 4). Lowering every comparison respondent's scores by the average difference predicted that 8.3% would have absent imagery in at least one sense. Lowering and also spreading out the scores predicted 12.0%. The real figure, 20.0%, was higher than both (bootstrap interval 13.3% to 26.7%). Both imaginary groups also predicted a smaller difference at the 10th percentile than we observed. They predicted a clear difference among people with imagery in every sense (-0.38 and -0.28 standard deviations), where the real difference was close to zero. In New et al.'s survey, then, the data point to a separate subgroup of autistic respondents who report no imagery in one or more senses.

Arnold et al.'s data. Absent imagery in at least one sense was reported by 9.8% of autistic adults and 4.6% of comparison adults (Fisher's exact test, p = .015). At the 10th percentile the autism group scored 0.67 standard deviations lower (95% CI -1.07 to -0.28); at the median the difference was only 0.04 (95% CI -0.23 to +0.15). Among people with imagery in every sense the difference was 0.07 (95% CI -0.22 to +0.08). Here, however, the shift + stretch version matched the real data well. It predicted that 8.9% would have absent imagery in at least one sense (real figure 9.8%), and a difference of 0.10 among people with imagery in every sense (real unadjusted figure 0.12). Only the difference at the 10th percentile was larger than predicted (real -0.75, 95% CI -1.08 to -0.41; predicted -0.39). In Arnold et al.'s data, therefore,

most of the pattern can be explained without a separate subgroup: autistic adults' scores were somewhat lower on average and more spread out, which by itself puts more of them at the bottom of the scale.

When we kept only the 242 autistic adults in Arnold et al.'s AQ-selected subgroup, a difference remained even among people with imagery in every sense (-0.19, 95% CI -0.37 to -0.02).

ADHD. In New et al.'s survey, 8.3% of the ADHD-only group reported absent imagery in at least one sense. This was close to the comparison group's 6.5% (Fisher's exact test, p = .49) and lower than the autism group's 20.0% (odds ratio 2.75, p = .024).

Table 3. Where the difference sits (exploratory)

Measure	New et al.	Arnold et al.
Absent imagery in at least one sense, autism vs comparison	20.0% vs 6.5% (Fisher p < .001)	9.8% vs 4.6% (Fisher p .015)
Extra senses with absent imagery, per person	+0.34 (+0.14 to +0.54)	+0.16 (+0.02 to +0.31)
Extra senses with the top score, per person	-0.03 (-0.21 to +0.15)	+0.00 (-0.19 to +0.19)
Difference at the 10th percentile (SD)	-1.27 (-1.66 to -0.88)	-0.67 (-1.07 to -0.28)
Difference at the 25th percentile (SD)	-0.56 (-0.81 to -0.30)	-0.27 (-0.53 to -0.02)
Difference at the 50th percentile (SD)	-0.19 (-0.40 to +0.02)	-0.04 (-0.23 to +0.15)
Difference at the 75th percentile (SD)	-0.09 (-0.32 to +0.13)	-0.02 (-0.21 to +0.16)
Difference at the 90th percentile (SD)	-0.15 (-0.37 to +0.07)	+0.14 (-0.05 to +0.34)
Difference among people with imagery in every sense (SD)	-0.03 (-0.21 to +0.14), n = 819	-0.07 (-0.22 to +0.08), n = 607

The first row compares percentages with Fisher's exact test. The other rows are differences between the autism and comparison groups, adjusted for age and sex, with 95% confidence intervals. Negative values mean the autism group scored lower. A percentile difference compares the groups at the same point in their range of scores: the 10th percentile compares the lowest-scoring 10% of each group, and the 50th percentile (median) compares the middle of each group. These were estimated with quantile regression. Each dataset uses its own six senses and is measured against its own comparison group; Arnold et al.'s sixth sense is bodily sensation rather than movement. Groups: New et al., 150 autism and 749 comparison (898 with age recorded); Arnold et al., 327 and 327. These analyses were not part of the registered plan.

Table 4. Observed autism group versus two whole-group models (exploratory)

Dataset and measure	Real autism group (95% CI)	Predicted by shift	Predicted by shift + stretch
New et al.: Absent imagery in at least one sense	20.0% (13.3% to 26.7%)	8.3%	12.0%
New et al.: Difference at the 10th percentile (SD)	-1.30 (-1.64 to -1.11)	-0.41	-0.76
New et al.: Difference at the median (SD)	-0.17 (-0.35 to +0.04)	-0.41	-0.39
New et al.: Difference among people with imagery in every sense (SD)	-0.03 (-0.21 to +0.14)	-0.38	-0.28
Arnold et al.: Absent imagery in at least one sense	9.8% (6.7% to 12.8%)	4.9%	8.9%
Arnold et al.: Difference at the 10th percentile (SD)	-0.75 (-1.08 to -0.41)	-0.23	-0.39
Arnold et al.: Difference at the median (SD)	-0.18 (-0.32 to -0.03)	-0.23	-0.19
Arnold et al.: Difference among people with imagery in every sense (SD)	-0.12 (-0.23 to -0.01)	-0.23	-0.10

This table asks whether a simple change to the comparison group's scores could reproduce the real autism group. Shift: every comparison respondent's score in each sense was lowered by the average difference between the groups in that sense. Shift + stretch: scores were also spread out to match the autism group's variability. Scores were rounded and kept within the questionnaire's range. If a prediction is close to the real value, a simple shift or spread is enough to explain the result; if not, a separate subgroup is more likely. For comparison, 6.5% (New et al.) and 4.6% (Arnold et al.) of the comparison groups had absent imagery in at least one sense. All values are unadjusted. Intervals come from resampling the autism group 2,000 times (bootstrap). Added after an internal review on 18 October 2026, after all other results were known.

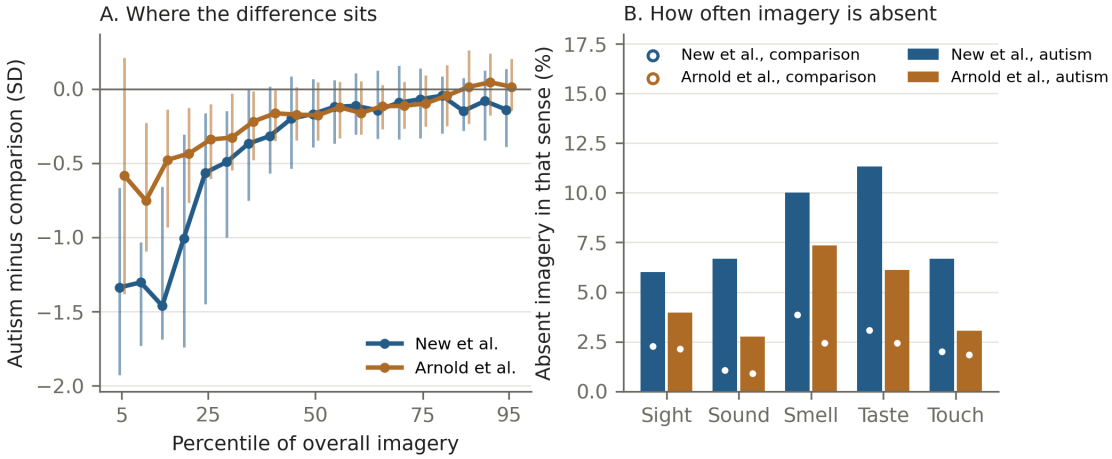


Figure 2. Where the difference lies (exploratory). Panel A shows the unadjusted difference in overall imagery between the autism and comparison groups at every 5th percentile, from the lowest scorers (left) to the highest (right), with 95% bootstrap intervals, in New et al.'s data (blue) and Arnold et al.'s data (orange). Among the lowest scorers the autism group scores much lower; from the median upwards the two groups are close. The intervals for the very lowest scorers are wide. Panel B shows how often imagery is absent, as the percentage of each group giving the lowest possible score in each sense. Bars show the autism groups and circles the comparison groups, in matching colours. Only the five senses that both questionnaires measure in the same way are shown.

Discussion

In a new sample and with a different questionnaire, our registered analysis replicated the three main findings of Arnold et al. (2026). Young adults who reported diagnosed or strongly suspected autism reported weaker imagery overall, and more of them reported no smell or taste imagery at all. The overall difference was larger than in Arnold et al.'s sample (-0.42 compared with -0.18 standard deviations). The two samples differ in age, in how autism was identified and in the questionnaire used, and we did not compare them statistically.

Absent smell and taste imagery was more common in the autism group, but the average gap in those senses was no larger than in sight (S1). In New et al.'s survey, absent imagery was more common in the autism group in every sense (Table S6). Smell and taste have the highest rates of absent imagery in both groups, so the gap in percentage points is largest there. The result is therefore not a problem specific to smell and taste. Absent imagery is more common

in autism across the senses, including sight (odds ratio 2.8, p = .054 after correction).

In these data, the difference did not extend to ADHD. The ADHD-only group was close to the comparison group both on overall imagery (-0.07 standard deviations) and on absent imagery (8.3% compared with 6.5%). The direct comparison of autism with ADHD was not statistically significant after correcting for the number of tests. Showing that the difference is specific to autism will need a larger ADHD group.

In New et al.'s survey, the difference comes mainly from a subgroup. One in five autistic respondents reported no imagery in at least one sense. A simple downward shift or a wider spread of everyone's scores could not reproduce this, and autistic respondents who reported some imagery in every sense scored like the comparison group. Arnold et al.'s data also showed more absent imagery in autism. There, however, most of the pattern could be explained by autistic adults' scores being a little lower and more spread out, without a separate subgroup. The two samples differ in two ways that may matter. New et al.'s respondents were younger, and their autism group included people who strongly suspected they were autistic but had not been diagnosed.

When a subgroup drives a difference, the group average is a poor summary. In New et al.'s survey, a difference of 0.4 standard deviations is too small to describe the one in five autistic respondents with absent imagery, and too large to describe everyone else. Some earlier studies already report the percentage of people who score as aphantasic or report absent imagery (King et al., 2024; Arnold et al., 2026). Comparing those percentages with what a simple shift in scores would predict, as in Table 4, shows whether the difference lies in a subgroup. A subgroup with absent imagery would also fit the finding that people with aphantasia report more autistic traits (Dance, Jaquiery et al., 2021). Two exploratory results bear on this. Among Arnold et al.'s autistic adults, those with absent imagery scored higher on the AQ questions that are not about imagination (d = 0.35, p = .046). This suggests that the link between absent imagery and autistic traits is not just a result of the AQ asking about

imagination. Our reanalysis of Dance, Ward and Simner's (2021) data is less clear. There, higher AQ scores went with weaker imagery among people without aphantasia (rho = -0.27), but not among people with aphantasia (rho = -0.01; Table S7). Finally, in Arnold et al.'s AQ-selected subgroup, a small difference remained even among people with imagery in every sense. Because some AQ questions ask about imagination, choosing people by their AQ score also tends to choose people with weaker imagery (Bouyer et al., 2025).

Limitations

Group membership rests on what people reported, and New et al.'s question combined diagnosis with strong suspicion, so the autism group probably includes some people without a diagnosis. Both datasets come from online questionnaires completed at a single time point, and New et al.'s sample is young and UK-based. The two questionnaires differ, and our registered plan wrongly treated Arnold et al.'s sixth sense, bodily sensation, as movement. Leaving that sense out changed the results little, but this check was added afterwards. The files available to us do not describe how Arnold et al. recruited their comparison group, or whether anyone took part in both surveys, so we did not compare the two datasets statistically. All the analyses of where the difference lies, including the shift benchmark, were added after the registered results were known. The shift benchmark is also simple: it keeps the shape of the comparison group's scores and changes only their average and spread. Some logistic models relied on the simplest kind of confidence interval, which can be inaccurate when one category is very small. Finally, an imagery score cannot be used to identify autism in an individual.

Conclusion

A preregistered test in a new sample replicated Arnold et al.'s finding that autistic adults report weaker imagery overall and are more likely to report no smell or taste imagery. The difference was no larger for smell and taste than for the other senses, and it was not seen in ADHD. In the newer sample, the difference came mainly from a minority of autistic adults who report no imagery at all in one or more senses. The next step is a registered test of this pattern in a third sample, ideally with clinically confirmed diagnoses and a measure of autistic traits that does not ask about imagination.

Supplementary material

Tables S1–S8 and Figure S1 are provided as supplementary material.

CRediT authorship contribution statement

Ivan Habib: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

[CONFIRM] The author declares no competing interests.

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Data and code availability

The data analysed are publicly available: New et al. (2026) at https://osf.io/2bdn6; Arnold et al. (2026) at https://doi.org/10.48610/57f325b; Dance, Ward and Simner (2021) at https://osf.io/h24vm. The preregistration is at https://osf.io/87udm. Analysis code and aggregate outputs are at https://github.com/i-habib/aphantasia/tree/main/projects/P5_asd_adhd_replication. No participant-level data are redistributed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[CONFIRM TOOLS AND VERSIONS] During the preparation of this work the author used Claude (Anthropic; Claude Opus 5.5) and OpenAI Codex [model] in order to write and check analysis code, run numerical checks, and draft and edit text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Supplementary material

Table S1. Participants and raw scores

Measure	Comparison	Autism	ADHD-only
Retained participants	749	150	84
Age (years)	25.08 (3.57)	25.01 (3.37)	25.19 (3.37)
Overall imagery (comparison SD)	0.00 (1.00)	-0.43 (1.40)	-0.07 (1.16)
Sight score (5-35)	24.57 (6.23)	22.22 (8.18)	24.05 (6.75)
Sound score (5-35)	26.85 (5.71)	24.48 (8.30)	26.52 (6.20)
Smell score (5-35)	23.05 (7.15)	20.89 (8.87)	22.62 (7.78)
Taste score (5-35)	25.49 (6.96)	22.75 (9.23)	25.10 (7.57)
Touch score (5-35)	24.91 (6.49)	22.83 (7.98)	24.33 (7.32)
Movement score (5-35)	26.08 (6.29)	24.29 (8.30)	26.13 (6.84)

Means (standard deviations), except participant counts. Ages: Comparison 18-32 years (n = 748); Autism 18-30 years (n = 150); ADHD-only 19-30 years (n = 84). Each sense score is the sum of five 1-7 ratings (range 5-35). Movement is New et al.'s sixth sense. Sex is shown in Table 1. The archive codes sex as 0, 1 and O; we read 0 as female and 1 as male because the coded totals across New et al.'s two studies (992, 952 and 8) match their published counts of female, male and other respondents in that order.

Table S2. Registered primary tests, full precision

Test	Estimate	95% CI	One-sided p	Supported
H1. Overall imagery, autism vs comparison	-0.422	-0.658 to -0.186	0.000232	Yes
H2a. Absent smell imagery	2.691	1.420 to 5.099	0.0012	Yes
H2b. Absent taste imagery	3.812	2.001 to 7.262	2.35e-05	Yes

The first estimate is a difference in comparison-group standard deviations; the others are odds ratios. The file supplement_registered_estimates.csv stores two-sided p values for every model; the one-sided p values above are the registered tests.

Table S3. Registered secondary tests

Test	Definition	Two-sided p	Holm-adjusted p
S1(a)	Autism: smell/taste difference minus sight difference	0.76	1
S1(b)	Autism: difference varies across the six senses	0.501	1
S2	Autism minus ADHD-only, overall score	0.0373	0.149
S3	Absent visual imagery, autism vs comparison	0.0108	0.0541
S4	ADHD-only: smell/taste difference minus sight difference	0.797	1

The ADHD-only versus comparison difference in the three-group model was -0.065 SD (95% CI -0.325 to +0.195). S1 and S4 use one row per person per sense (5,388 rows for 898 people in S1; 4,992 rows for 832 people in S4).

Table S4. Registered sensitivity analyses and Arnold et al. reanalysis

Analysis	Estimate	95% CI	Units	n
New: autism group without co-reported ADHD	-0.389	-0.651 to -0.128	SD; people	852
New: raw six-sense mean	-2.216	-3.456 to -0.976	Points on 5-35 scale; people	898
New: ages 18-30 only	-0.421	-0.657 to -0.185	SD; people	895
New: no age/sex adjustment	-0.428	-0.664 to -0.191	SD; people	899
New: absent smell, without co-reported ADHD	2.871	1.406 to 5.865	Odds ratio; people	852
New: absent taste, without co-reported ADHD	3.910	1.904 to 8.031	Odds ratio; people	852
New: sight difference, age/sex effects allowed to vary by sense	-0.380	-0.602 to -0.158	Sight SD; person-sense rows	5388
Arnold: overall imagery, six senses	-0.180	-0.359 to -0.002	SD; people	654
Arnold: sight difference in the per-sense model	-0.134	-0.307 to 0.039	Sight SD; person-sense rows	3924

In the per-sense models, sight is the reference sense, so the group coefficient is the difference for sight only. New et al.'s 5,388 rows are 898 people; Arnold et al.'s 3,924 rows are 654 people. All rows, including interaction terms, are in supplement_registered_estimates.csv.

Table S5. Five-sense check (exploratory)

Dataset	Difference (SD)	95% CI	Two-sided p	n
New et al.	-0.435	-0.669 to -0.201	0.000275	898
Arnold et al.	-0.199	-0.377 to -0.021	0.0286	654

Uses sight, sound, smell, taste and touch only, standardised against each dataset's comparison group, with the same age/sex-adjusted model as H1. Specified in the deviations log on 17 October 2026, after the registered results were known.

Table S6. Lowest and highest scores by sense (exploratory)

Sense	New: absent, autism / comparison	New: top score, autism / comparison	Arnold: absent, autism / comparison	Arnold: top score, autism / comparison
Sight	6.0% / 2.3%	3.3% / 4.3%	4.0% / 2.1%	8.3% / 7.3%
Sound	6.7% / 1.1%	8.7% / 9.7%	2.8% / 0.9%	11.6% / 10.7%
Smell	10.0% / 3.9%	3.3% / 3.5%	7.3% / 2.4%	3.4% / 4.6%
Taste	11.3% / 3.1%	6.7% / 8.7%	6.1% / 2.4%	5.2% / 6.1%
Touch	6.7% / 2.0%	6.0% / 5.9%	3.1% / 1.8%	10.4% / 9.8%
Movement (New) / bodily sensation (Arnold)	8.0% / 2.1%	9.3% / 8.7%	2.4% / 1.5%	5.5% / 4.9%

Absent: the lowest possible score in that sense (5 of 35 in New et al.; 0 of 50 in Arnold et al.). Top score: the highest possible score (35; 50). New et al.: 150 autism, 749 comparison. Arnold et al.: 327 each.

Table S7. All other exploratory analyses

Every exploratory analysis run on these data after the registered results were known, including those not discussed in the main text. Differences are autism minus comparison, adjusted for age and sex, with 95% confidence intervals, unless stated otherwise.

Analysis	Result	n
New: overall difference among people who repeated all five activities in the attention check	-0.42 (-0.68 to -0.15) SD	702
New: emotion imagery (not in the overall score)	-0.43 (-0.65 to -0.21) SD, Holm p < .001	898
New: voice imagery	-0.27 (-0.50 to -0.03) SD, Holm p .026	898
New: inner-body imagery	-0.37 (-0.59 to -0.16) SD, Holm p .001	898
New: overall difference, also adjusted for current treatment	-0.42 (-0.66 to -0.18) SD	898
New: overall difference, also adjusted for education	-0.41 (-0.65 to -0.17) SD	889
New: empathy self-rating (1-7), autism minus comparison	-0.39 (-0.67 to -0.12) points	898
New: overall difference, also adjusted for the empathy rating	-0.34 (-0.57 to -0.10) SD	898
New: absent imagery in at least one sense, ADHD-only vs comparison	8.3% vs 6.5%, Fisher p .493	833
New: absent imagery in at least one sense, autism vs ADHD-only	OR 2.75, Fisher p .024	234
Arnold, AQ 26 or more only: absent imagery in at least one sense	11.6% vs 4.6%	569
Arnold, AQ 26 or more only: median difference	-0.20 (-0.42 to +0.01) SD	569
Arnold, AQ 26 or more only: difference among people with imagery in every sense	-0.19 (-0.37 to -0.02) SD	526
Arnold, autistic adults: AQ without imagination items, absent imagery in any sense vs none	30.0 vs 27.4, d = 0.35, Mann-Whitney p .046	327
Both datasets pooled: difference among people with imagery in every sense	-0.05 SD (-0.17 to +0.06); two one-sided tests against +/-0.2 SD, p .006	1426
Dance et al. (2021), people without aphantasia: AQ vs six-sense imagery	Spearman rho = -0.27, p .001	138
Dance et al. (2021), people with aphantasia: AQ vs six-sense imagery	Spearman rho = -0.01, p .885	164
Dance et al. (2021), general population: AQ vs visual imagery (VVIQ)	Spearman rho = 0.04, p .749	83

Why these analyses were run. Attention check: to see whether careless responding explains the difference. Emotion, voice and inner-body imagery: the three kinds of imagery New et al. measured that are not in the overall score. Treatment and education: possible confounders recorded in the survey. Empathy self-rating: the survey's only non-imagery 1-7 rating, used to check whether autistic respondents simply rate themselves lower on any 1-7 item; adjusting for it reduced the difference from -0.42 to -0.34. AQ 26 or more: Arnold et al.'s own selection. AQ without imagination items: whether

absent imagery goes with autistic traits beyond the items that ask about imagination (Bouyer et al., 2025). Dance, Ward and Simner (2021): a public dataset with AQ and six-sense imagery in people with and without aphantasia (OSF h24vm; their AQ total omits item 3). The pooled estimate uses fixed-effect inverse-variance weights across two questionnaires whose sixth senses differ, and it selects people by their scores; the +/-0.2 SD bound is a conventional 'small' effect, chosen after the results were known.

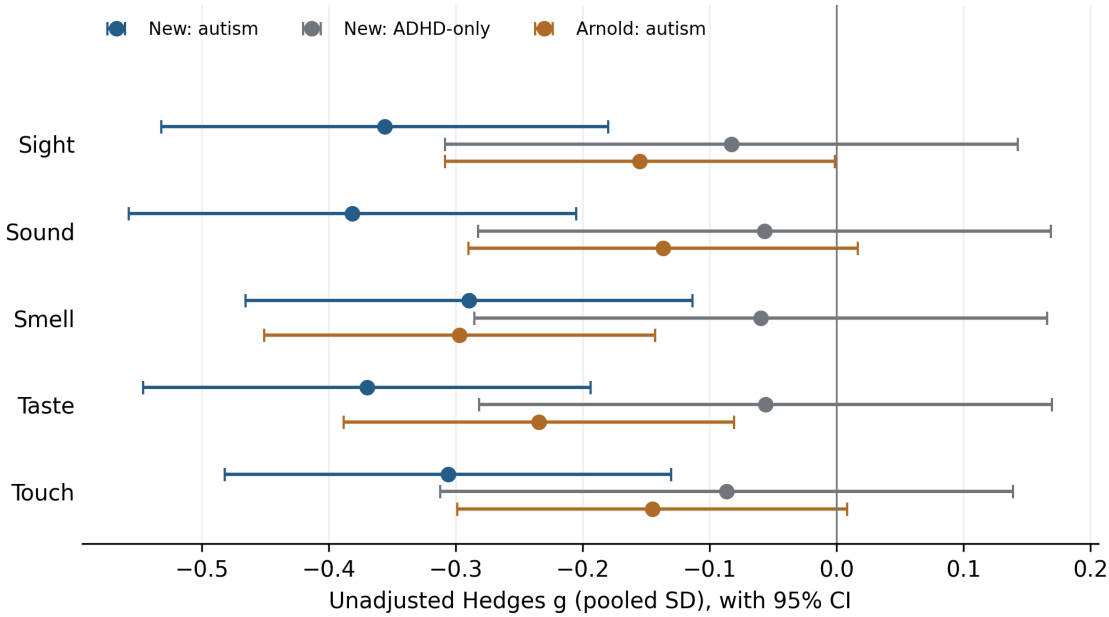


Figure S1. Unadjusted differences in each of the five senses measured in both questionnaires (Hedges g with pooled standard deviations and 95% confidence intervals). This scale differs from the adjusted overall score in the main text. These are descriptive, not additional tests.

Reproduction and checks

All 156 registered estimate rows were reproduced exactly in a separate run (largest absolute difference 0.0). The frozen plan's SHA-256 is ce102d64229b6a4efb2a7c646048f17c5ab338829f73bed4121bcc1deb147c14, and the original output files were unchanged.

The overall-score regression was rebuilt independently with statsmodels (OLS, HC3, t inference) and agreed. Holm adjustment was rechecked with statsmodels. The five Firth fits were refitted in R 4.6.1 with brglm2 1.1.1 (mean bias reduction). Coefficients, covariance

matrices, Wald intervals and p values agreed: the largest coefficient difference was 3.7e-07 and the largest covariance difference 8.5e-07. To converge with the small 'other' sex category, the R fit needed an intercept-only start and step size 0.25.

The code, run order and checks are documented in the repository; no participant-level rows are written.

Source measures

New et al.: OSF project 2bdn6. The data file for Study 2 is named Exp3.1020.txt in the archive. The questionnaire (Study2.Questionnaires.pdf in the archive) is the 45-item QMI + VocEmo, New et al.'s extension of the Questionnaire upon Mental Imagery with vocal and emotional items. The six senses used here are its sight, sound, smell, taste, touch and movement scales. Its emotion, voice and inner-body scales are not in the overall score. New et al.'s attention rule keeps people who repeated at least three of their five favourite activities.

Arnold et al.: the public workbook labels its six PSI-Q columns visual, auditory, smell, taste, touch and bodily sensation. All 327 autistic records in Arnold et al.'s analysis code match the workbook on imagery scores, AQ and age, and all have the maximum attention score (13). How the comparison group was recruited is not documented in these files.

Table S8. Deviations from the registered plan

Date	Change	Effect
10 Oct 2026	The plan gave the effect Arnold et al.'s study had a 33% chance of detecting (used to judge a null H1) as about 0.12 SD; the registered code computed 0.139 SD.	None: only used if H1 had failed.
10 Oct 2026	Firth regression used for H2a, H2b, S3 and their sensitivity analyses, as the plan allowed, because the 'other' sex category perfectly predicted the outcome.	None; reported as planned.
11 Oct 2026	The plan's SHA-256 for the New et al. data file was the archive's fingerprint, not the file's. Label corrected.	None; same data.
17 Oct 2026	The plan treated Arnold et al.'s sixth sense (bodily sensation) as movement. Registered results kept as run; five-sense check added.	Registered six-sense results are not matched across questionnaires.
17 Oct 2026	Registered analysis reproduced in a separate directory; Firth fits checked against R.	Numerical check only; no discrepancies.
18 Oct 2026	Shift benchmark (Table 4) added after an internal review.	Exploratory.

The full deviations log is in the repository alongside the registered plan.

Comparing the two datasets

The registered six-sense differences were -0.42 SD (New) and -0.18 SD (Arnold). Their difference, -0.24 SD, had a two-sided p of .11 under a normal approximation. This comparison treats the samples as independent and the two composites as comparable; neither is established, so we do not interpret it.